One-Pager Case Study :- Lead Generation Automation (n8n)

Project Title:

- Automated Lead Generation & Enrichment Workflow.

Tools Used:

- n8n, Apollo API, Google Drive, Google Sheets, JavaScript Functions.

1. Problem / Need

The sales team was performing manual research to collect and enrich lead data. This process was time-consuming, inconsistent, and often resulted in incomplete or duplicate information. There was a need for an automated system that could process lead lists, enrich them with verified data, and store them in a clean and structured format.

2. Solution (Workflow Approach)

- Google Drive Trigger: The workflow triggers automatically when a new lead sheet is uploaded.
- File Extraction: The sheet is downloaded and transformed into structured lead records.
- Data Cleaning: JavaScript nodes were used to trim spaces, normalize text, and handle missing values.
- Apollo API Integration: Each lead is enriched with verified business email, LinkedIn profile, company details, and location.
- Final Output: The cleaned and enriched lead data is appended into a Google Sheet in a standardized format.

3. Result / Outcome

Before Automation	After Automation
Manual lead research and data entry	Fully automated workflow
1–2 hours per sheet	3–5 minutes per sheet
Inconsistent and incomplete records	Clean and enriched data

4. Key Learning

- Workflow automation significantly reduces manual work.
- API-based enrichment improves data quality and usability.
- A structured workflow ensures scalability and reliability.